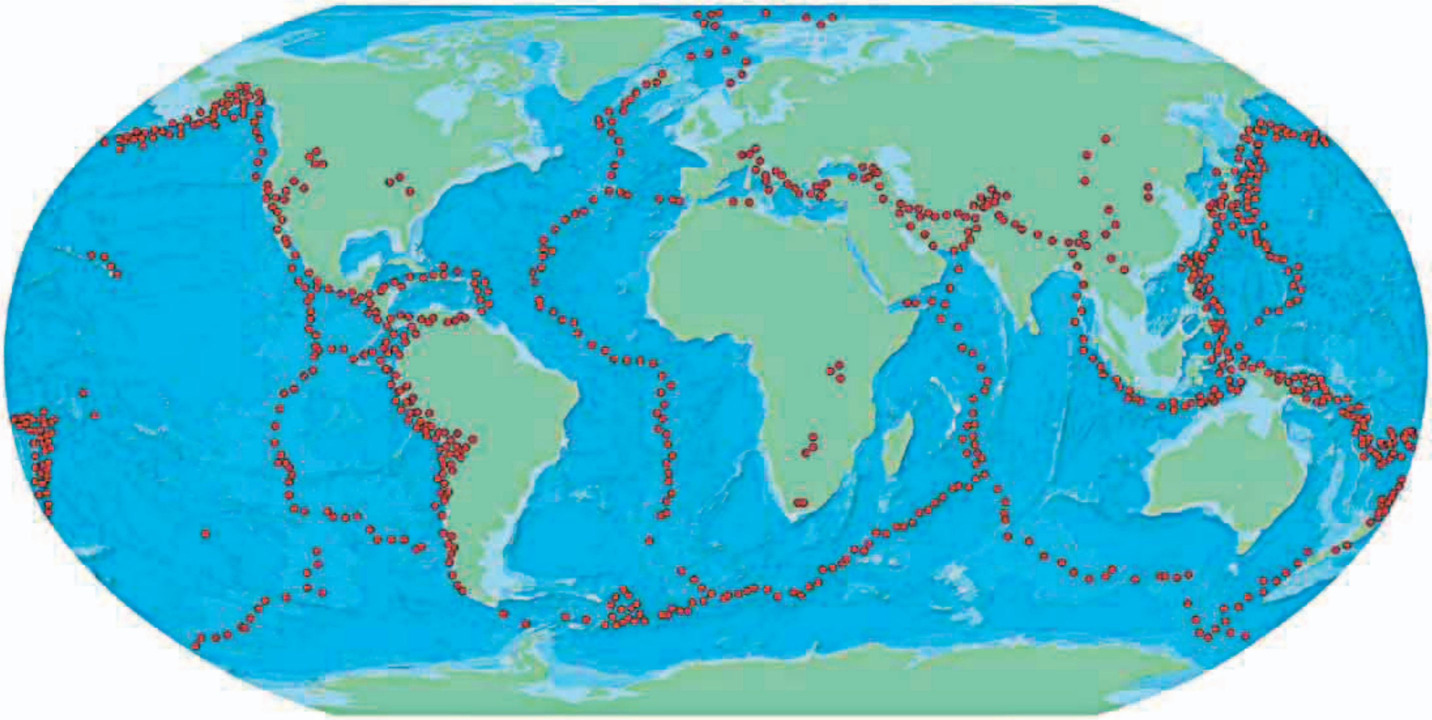
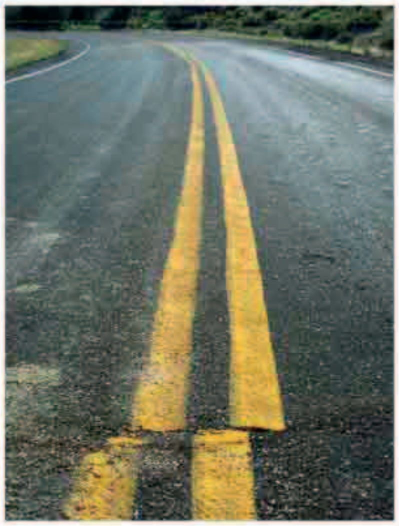
Earthquakes

Most earthquakes occur on the plate boundaries of Earth's crust, where the plates are moving relative to one another. This movement may cause just minor tremors, but if the plates lock together, the strain builds up until something snaps. It is this sudden release of tension that causes a big, destructive earthquake.

EARTHQUAKE ZONES

In the 1960s, the United States used a network of seismometers (instruments that record motions in the ground) to monitor nuclear bomb tests. The instruments also recorded all the earthquakes worldwide, and revealed that most of them (shown in red below) occur in narrow bands along the tectonic plate boundaries.





SLIPPING AND SLIDING

The plates of the crust are always moving, but very slowly. As they move, they slip against each other at plate boundaries. When this occurs at a steady rate, it causes gradual creeping movement that can crack road surfaces and move fences out of line. It also triggers frequent small earth tremors—often so small that no one notices them.

◂ MINOR SHIFT
This crack in the road was caused by creeping movement at or near the boundary between two moving plates.

FACT!

During the 1964 Alaska earthquake, the Pacific floor slid 66 ft (20 m) beneath Alaska in just three minutes, and lifted an old shipwreck out of the water.